

Solution Requirements

Date	15 March 2026
Team ID	
Project Name	Smart Lender : AI/ML-Based Loan Approval Prediction System

Step 1: Functional Requirements (FR)

S.NO	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	Users (customers and bank officers) must securely log in using username/password and OTP verification.	High
2	Authorization levels	Customers can apply and track loans. Bank officers can review, approve, or reject applications. Admin manages users and system settings.	High
3	External interfaces	The system connects to Credit Bureau APIs, Aadhaar/e-KYC verification, Email/SMS notification services, and Banking APIs.	High
4	Transactions processing	The system receives loan applications, validates documents, calculates AI credit score, processes approval/rejection, and updates loan records.	High
5	Reporting	Generate loan application reports, approval/rejection reports, customer history, repayment	Medium

		reports, and AI prediction dashboards.	
6	Business rules, etc.	Loan approval depends on AI credit score, income, employment status, repayment history, and document verification.	High
7	Compliance to laws or regulations	The system must comply with RBI digital lending guidelines, KYC requirements, and data privacy regulations.	High
8	Other	Send SMS/Email notifications for application submission, approval, rejection, EMI reminders, and repayment alerts.	Medium

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	The system should process loan applications quickly with minimal delay.	Response time less than 2 seconds for normal requests; AI assessment completed within 30 seconds .
2	Scalability	The system should support increasing users and loan applications without performance degradation.	Supports 10,000+ concurrent users and 100,000+ loan records .
3	Security & Data Privacy	Protect customer information using encryption, secure authentication, and access control.	AES-256 encryption , HTTPS, Role-Based Access Control (RBAC), Multi-Factor Authentication (MFA).
4	Reliability & Availability	The system should remain operational	

		with minimal downtime and recover from failures.	99.9% uptime , automatic backup and recovery.
5	Usability & Accessibility	Provide a simple, responsive, and accessible interface for all users.	Mobile-friendly UI, supports WCAG 2.1 AA accessibility standards.
6	Other	Maintainability and portability for future updates and deployment.	Modular architecture, cloud deployment support, easy software updates.